

Theoretical pb. (hardest)

5.) orthog. refl + translation $\Rightarrow$ reflection

$$\pi: \langle m, x \rangle = c.$$

$$\text{Ref}_{\pi}^{\perp}(x) = x - 2 \cdot \frac{\langle m, x \rangle - c}{\langle m, m \rangle} m$$

$$v \parallel m \Rightarrow v = \lambda m.$$

$$\langle v, m \rangle = \langle \lambda m, m \rangle = \lambda \langle m, m \rangle \Rightarrow \lambda = \frac{\langle v, m \rangle}{\langle m, m \rangle}.$$

$$\Rightarrow v = \lambda m = \frac{\langle v, m \rangle}{\langle m, m \rangle} m.$$

Translation function: $T_v(y) = y + v$

$$\begin{aligned} \Phi(x) &= (T_v \circ \text{Ref}_{\pi}^{\perp})(x) = T_v(\text{Ref}_{\pi}^{\perp}(x)) = \left(x - 2 \cdot \frac{\langle m, x \rangle - c}{\langle m, m \rangle} m \right) + v = \\ &= \left(x - 2 \cdot \frac{\langle m, x \rangle - c}{\langle m, m \rangle} m \right) + \frac{\langle v, m \rangle}{\langle m, m \rangle} m = x + \frac{-2(\langle m, x \rangle - c) + \langle v, m \rangle}{\langle m, m \rangle} m \\ &= x - 2 \cdot \frac{\langle m, x \rangle - (c + \frac{1}{2} \langle v, m \rangle)}{\langle m, m \rangle} m \end{aligned}$$

$$\Phi(x) = x - 2 \cdot \frac{\langle m, x \rangle - c'}{\langle m, m \rangle} m$$

$$\Rightarrow \pi': \langle m, x \rangle = c' = c + \frac{1}{2} \langle v, m \rangle.$$

! (2) case II: $l_1 \parallel l_2 \Rightarrow$ same unit vector $m. \Rightarrow \boxed{C \cdot A = I_2}$

$$l_1: \langle m, x \rangle = c_1$$

$$\Phi(x) = Ax + b = \text{Ref}_{l_1}^{\perp} \rightarrow \det A = -1; A = A^{-1}$$

$$l_2: \langle m, x \rangle = c_2$$

$$\Psi(x) = Cx + d = \text{Ref}_{l_2}^{\perp} \rightarrow \det C = -1; C = C^{-1}$$

$$(\Phi \circ \Psi)(x) = \underbrace{CA}x + Cb + d.$$

$$\begin{aligned} (CA)^T &= (CA)^{-1} \\ \det(CA) &= \det C \cdot \det A = (-1) \cdot (-1) = 1 \Rightarrow \text{det. isometry} \end{aligned}$$

$$\bullet \text{ id. (1): } (C \Rightarrow) \begin{cases} CA = I_2 \\ Cb + d = 0 \end{cases} \Rightarrow C = A^{-1} = A \Rightarrow Ab + d = 0 \Rightarrow d = -Ab. \Rightarrow d = b.$$

$$b \perp \text{axis of } A \Rightarrow \underline{Ab = -b}$$

$$\Rightarrow \Phi(x) = Ax + b$$

$$\Psi(x) = Cx + d = Ax + b = \Phi(x) \Rightarrow \text{same reflection.}$$

$$\Rightarrow \text{id} \Leftrightarrow l_1 = l_2. \text{ (actually case I, i.g.)}$$

• translation (2): $T_v(x) = x + v, v \neq 0.$

$$l_1 \neq l_2 \Rightarrow \text{translation if } l_1 \parallel l_2, l_1 \neq l_2.$$

$$\bullet \text{ rot. (3)} \quad CA = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} = I_2 \Leftrightarrow \theta = 0^\circ \Rightarrow \text{identity. (impossible.)}$$

26) $\mathcal{C}: \frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0 \rightarrow$ ruled surface?

(=) Through any point $\in \mathcal{C}$, $\exists$ at least one line (= generator) that sits entirely inside $\mathcal{C}$.

parametrization of the line passing through O and P using param t :

$$\begin{cases} L(t) = (tx_0, ty_0, tz_0), t \in \mathbb{R}. \\ \mathcal{C}: \dots \end{cases}$$

$$\Rightarrow \left(\frac{tx_0}{a} \right)^2 + \left(\frac{ty_0}{b} \right)^2 + \left(\frac{tz_0}{c} \right)^2 = t^2 \underbrace{\left(\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} - \frac{z_0^2}{c^2} \right)}_0 = 0, (\forall) t.$$

$$\Rightarrow L(t) \in \mathcal{C}. \Rightarrow \mathcal{C} = \text{ruled surface.}$$

$$\Rightarrow L(t) = \text{generator through } P.$$

27) $\mathcal{C}: \dots = 0 \rightarrow (\forall) P \in \mathcal{C}$, generator is contained in the tg plane at P .

$$\begin{pmatrix} F(x_0, y_0, z_0) = \left(\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} - \frac{z_0^2}{c^2} \right) \\ \nabla F(x_0, y_0, z_0) = \left(\frac{2x_0}{a^2}, \frac{2y_0}{b^2}, -\frac{2z_0}{c^2} \right) \\ \vdots \end{pmatrix}$$

$\Downarrow$

$$\begin{cases} \nabla_P \mathcal{C}: \frac{x x_0}{a^2} + \frac{y y_0}{b^2} - \frac{z z_0}{c^2} = 0 \\ L(t) = (tx_0, ty_0, tz_0) \end{cases} \Rightarrow \frac{tx_0^2}{a^2} + \frac{ty_0^2}{b^2} - \frac{tz_0^2}{c^2} = 0$$

$$= t \underbrace{\left(\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} - \frac{z_0^2}{c^2} \right)}_0 = 0.$$

Theoretical problems section 2 (harder ones)

① Projection along v formula deduction?

line l , dir. vector v , point P .

Proof

Fix $K = (0, 3)$ frame of H^m .

Hyperplane H : $a_1 x_1 + \dots + a_m x_m + a_{m+1} = 0$

line l_P passing through P with dir. vector v : $l_P = \{P + tv : t \in \mathbb{R}\}$

$$l_P \cap H \Rightarrow a_1(p_1 + t v_1) + a_2(p_2 + t v_2) + \dots + a_m(p_m + t v_m) + a_{m+1} = 0.$$

$$P' = P_{H,v}(P) = P - \frac{a_1 p_1 + \dots + a_m p_m + a_{m+1}}{a_1 v_1 + \dots + a_m v_m} v$$

$$= P - \frac{a^T \cdot P + a_{m+1}}{a^T \cdot v} v$$

$$\boxed{a^T \cdot P \cdot v = v \cdot a^T \cdot P}$$

$$= P - \frac{a^T \cdot P \cdot v}{a^T \cdot v} - \frac{a_{m+1}}{a^T \cdot v} v$$

$$\langle a, v \rangle = a^T \cdot v = \langle v, a \rangle = v^T \cdot a$$

$$= P \left(I_m - \frac{v \cdot a^T}{v^T \cdot a} \right) - \frac{a_{m+1}}{v^T \cdot a} v$$

if ORTHONORMAL

$$\langle a, v \rangle = a^T v$$

$$\boxed{v \otimes a = v \cdot a^T}$$

$$\Rightarrow \left(I_m - \frac{v \otimes a}{\langle v, a \rangle} \right)$$

$$\Rightarrow P_{H,v}(P) = P \left(I_m - \frac{v \cdot a^T}{\langle v, a \rangle} \right) - \frac{a_{m+1}}{v^T \cdot a} v \quad (1)$$

$$\textcircled{2} \quad \boxed{P_{H^\perp}^\perp(P) = (I_3 - m \otimes m)P}, \quad |m| = 1.$$

$$(1) \Rightarrow P_{H^\perp}^\perp(P) = \left(I_m - \frac{a \otimes a}{|a|^2} \right) P - \frac{a_{m+1}}{|a|^2} \cdot a \quad (1)$$

$$\langle a, a \rangle = \|a\|^2$$

Hyperplane: H : $a_1 x_1 + \dots + a_m x_m + a_{m+1} = 0$ K

H : $a^T P + a_{m+1} = 0$ (gen. hyp. eq.)

our hyp: $m^T P = 0$

$$\Rightarrow a^T P + a_{m+1} = m^T P$$

$$\Rightarrow \begin{cases} m^T = a^T \\ \underline{a_{m+1} = 0} \end{cases}$$

$$\Rightarrow \underline{m = a} \xrightarrow{(1)} P_{H^\perp}^\perp(P) = \left(I_m - \frac{m \otimes m}{|m|^2} \right) P - \frac{0}{\dots}$$

$|m| = 1$

$$\Rightarrow P_{H^\perp}^\perp(P) = (I_m - m \otimes m)P.$$

③ $\text{Ref}_{\frac{1}{2}}^{\perp}(P) = (I_3 - 2 m \otimes m) P$, $|m|=1$

$\text{Pr}_{\frac{1}{2}}^{\perp}(P) = (I_3 - m \otimes m) P$

$(m \otimes m) P = \langle m, P \rangle m$

$P = I_3 \cdot P = \underbrace{(I_3 - m \otimes m) P}_{\text{part in plane}} + \underbrace{(m \otimes m) P}_{\text{normal part}}$

$\text{Ref}_{\frac{1}{2}}^{\perp}(P) = (I_3 - m \otimes m) P - (m \otimes m) P = (I_3 - 2 m \otimes m) P$

⑤ orthog. reflection + translation $\Rightarrow$ reflection
 $(-1) \cdot (1) = (-1)$

Π plane in $\mathbb{R}^3$

$\Pi: \langle m, x \rangle = c$

$v \parallel m \Rightarrow v = \lambda m$

$T_v \circ \text{Ref}_{\frac{1}{2}}^{\perp} = \text{reflection in } \bar{u}: \langle m, x \rangle = c + \frac{1}{2} \langle v, m \rangle$

Proof:

plane $\langle m, x \rangle = c \Rightarrow \text{Ref}_{\frac{1}{2}}^{\perp}(x) = x - 2 \cdot \frac{\langle m, x \rangle - c}{\langle m, m \rangle} m$ ($m = \text{unit}$)
 $\Rightarrow \text{Ref}_{\frac{1}{2}}^{\perp}(x) = x - 2 \cdot (\langle m, x \rangle - c) m$
 (orthogonal reflection)

$T_v = x + v$ (translation)

$(T_v \circ \text{Ref}_{\frac{1}{2}}^{\perp})(x) = x - 2 \cdot (\langle m, x \rangle - c) \cdot m + v$

$v \parallel m \Rightarrow v = \lambda m = \langle v, m \rangle m$
 $|m|=1$

$\langle v, m \rangle = \langle \lambda m, m \rangle = \lambda \langle m, m \rangle = \lambda \cdot |m|^2 = \lambda$

$(T_v \circ \text{Ref}_{\frac{1}{2}}^{\perp})(x) = x - 2 \cdot (\langle m, x \rangle - c) \cdot m + \langle v, m \rangle m$
 $= x - 2 \cdot (\langle m, x \rangle - c - \frac{1}{2} \langle v, m \rangle) m$
 $= x - 2 \cdot (\langle m, x \rangle - (c + \frac{1}{2} \langle v, m \rangle)) m$

$\langle v, m \rangle = \langle \lambda m, m \rangle = \lambda \langle m, m \rangle = \lambda \cdot |m|^2 = \lambda \cdot 1 = \lambda$

$\Rightarrow v = \lambda m = \frac{\langle v, m \rangle}{\langle m, m \rangle} m$

$\bar{u}: \langle m, x \rangle = c + \frac{1}{2} \langle v, m \rangle$

$\Phi(x) = T_v(\text{Ref}(x)) = \left(x - 2 \cdot \frac{\langle m, x \rangle - c}{\langle m, m \rangle} m \right) + v = \left(x - 2 \cdot \frac{\langle m, x \rangle - c}{\langle m, m \rangle} m \right) + \frac{\langle v, m \rangle}{\langle m, m \rangle} m$

Rotation with θ around O after reflection in x -axis $\Rightarrow$ reflection in $y = \tan \frac{\theta}{2} x$

• refl. in x -axis: $(x, y) \mapsto (x, -y)$

$$H = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

• Rot. with θ around O : $R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

• Rot. after reflection: $R_\theta \circ H = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} \cos \theta & +\sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}$

• refl. in line through O , making angle θ with the x -axis.

$$\begin{pmatrix} \cos 2\alpha & \sin 2\alpha \\ \sin 2\alpha & -\cos 2\alpha \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} = R_\theta \circ H$$

$$\Rightarrow 2\alpha = \theta \Rightarrow \alpha = \frac{\theta}{2}$$

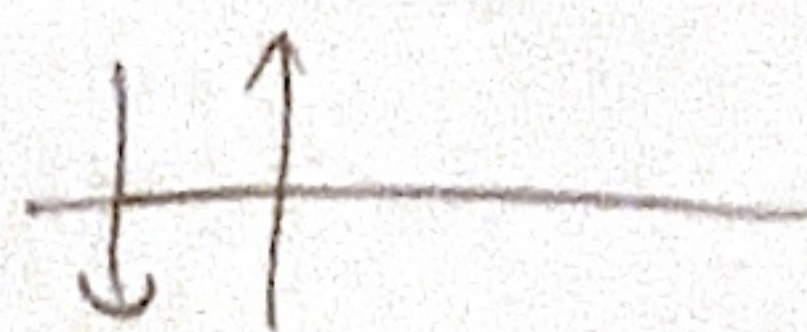
$$\boxed{y = \tan(\alpha) \cdot x}, \quad \alpha = \frac{\theta}{2}$$

$$y = \tan \frac{\theta}{2} \cdot x$$

⑫ refl. in line 1 $\circ$ refl. in line 2

Case I: $l_1 = l_2$

$$\Phi = \text{Ref}_{l_1}^\perp \circ \text{Ref}_{l_2}^\perp \Rightarrow \text{id.}$$



Case II: $l_1 \perp l_2$

$$\Rightarrow \det A = -1; \det C = -1.$$

$$\Phi(x) = Ax + b = \text{Ref}_{l_1}^\perp$$

$$\Psi(x) = Cx + d = \text{Ref}_{l_2}^\perp$$

$$\Rightarrow \Psi(\Phi(x)) = C(Ax + b) + d = \underbrace{CA^x}_{\text{lin. trans.}} + \underbrace{Cb + d}_{\text{trans.}}$$

$$\Psi \circ \Phi \text{ Dir. isometry because } \det(A \cdot C) = \det A \cdot \det C = (-1) \cdot (-1) = 1, (CA)^T = (CA)^{-1}.$$

id.

trans?

rot?

$$\text{identity } (\Leftrightarrow) \begin{cases} CA = I_2 \\ Cb + d = 0 \end{cases} \Rightarrow C = A^{-1} \Rightarrow A = C = A^{-1} \Rightarrow \begin{cases} \Phi(x) = Ax + b \\ \Psi(x) = Ax + d \end{cases}$$

$$\Psi \circ \Phi = A^2 x + \underbrace{Ab + d}_0 \Rightarrow Ab + d = 0 \Rightarrow$$

$$\boxed{b \perp \text{axis of } \Psi} (\Leftrightarrow) \Phi(x) = Ax + b \text{ reflection.}$$

Case III: $\sin \theta = 1$

$\Psi \circ \Phi$ dir. isom. with 1 fixed point

5

Euler-Rodrigues

$u = \text{unit vector} \Rightarrow |u| = 1$

$\theta \in \mathbb{R}$

Rot. with angle θ and axis u : $\text{Rot}_{u,\theta}(x) = \cos\theta x + \sin\theta(u \times x) + (1 - \cos\theta)\langle u, x \rangle u$

$$x = x_{\parallel} + x_{\perp}$$

$$x_{\parallel} = \langle u, x \rangle u$$

$$x_{\perp} = x - x_{\parallel} = x - \langle u, x \rangle u$$

$$u \times x \perp u \\ \perp x$$

$$\text{Rot}_{u,\theta}(x_{\parallel}) = \text{Rot}_{u,\theta}(\langle u, x \rangle u) = \langle u, x \rangle u$$

$$\text{Rot}_{u,\theta}(x_{\perp}) = \text{Rot}_{u,\theta}(x - \langle u, x \rangle u) = \cos\theta x_{\perp} + \sin\theta(u \times x)$$

$$\text{Rot}_{u,\theta}(x) = \text{Rot}_{u,\theta}(x_{\parallel}) + \text{Rot}_{u,\theta}(x_{\perp})$$

$$= \langle u, x \rangle u + \cos\theta x_{\perp} + \sin\theta(u \times x)$$

$$= \langle u, x \rangle u + \cos\theta(x - \langle u, x \rangle u) + \sin\theta(u \times x)$$

$$= \cos\theta x + \sin\theta(u \times x) + (1 - \cos\theta)\langle u, x \rangle u.$$